CYCLE SALES & CUSTOMER ANALYSIS





PRESENTATION 2026



WELCOME TO OUR COMPANY

We are a leading bicycle manufacturing company focused on delivering high-quality, reliable, and innovative bicycles. Our products are designed for everyday commuting, fitness, adventure, and sustainable transportation. With a strong focus on quality, customer satisfaction, and continuous improvement, we aim to make cycling more accessible and enjoyable for everyone.

VISION & MISSION

OUR VISION

To become a trusted and innovative bicycle brand that promotes healthy, active, and sustainable lifestyles worldwide.

OUR MISSION

To provide high-quality, affordable, and reliable bicycles while delivering excellent customer satisfaction and contributing to a cleaner and healthier environment.

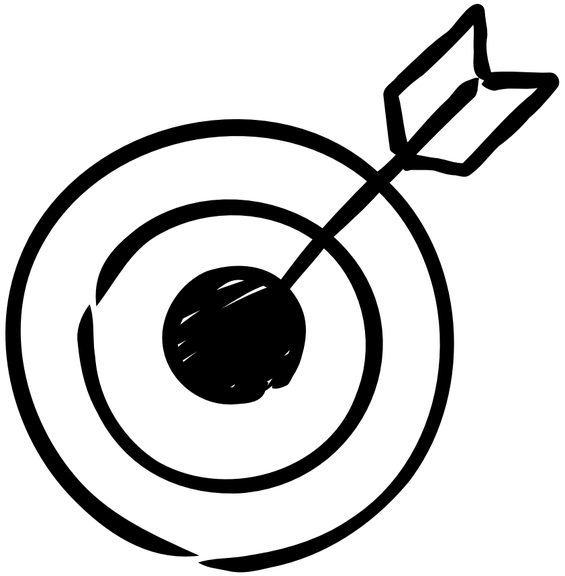
WHY SHOULD YOU CHOOSE US?

We are committed to delivering high-quality bicycles that combine performance, comfort, innovation, and value. Our focus on quality manufacturing and customer satisfaction helps us build products that customers can rely on.



- **Quality Products**
- **Timely Delivery**
- **Cost Efficiency**
- **Customer Focused**

BUSINESS OBJECTIVE



The objective of this project is to analyze bicycle sales data using SQL and identify important business trends to support better decision-making.

- **Sales Performance**
- **Product Performance**
- **Customer Analysis**
- **Business Growth**

DATASET OVERVIEW

CUSTOMER TABLE

	customer_key [PK] integer	customer_id integer	customer_number character varying (20)	country character varying (50)	marital_status character varying (20)	gender character varying (10)	birthdate date	create_date date	name character varying (100)
1	1	11000	AW00011000	Australia	Married	Male	1971-10-06	2025-10-06	Jon Yang
2	2	11001	AW00011001	Australia	Single	Male	1976-05-10	2025-10-06	Eugene Huang
3	3	11002	AW00011002	Australia	Married	Male	1971-02-09	2025-10-06	Ruben Torres
4	4	11003	AW00011003	Australia	Single	Female	1973-08-14	2025-10-06	Christy Zhu
5	5	11004	AW00011004	Australia	Single	Female	1979-08-05	2025-10-06	Elizabeth Johnson
6	6	11005	AW00011005	Australia	Single	Male	1976-08-01	2025-10-06	Julio Ruiz
7	7	11006	AW00011006	Australia	Single	Female	1976-12-02	2025-10-06	Janet Alvarez
8	8	11007	AW00011007	Australia	Married	Male	1969-11-06	2025-10-06	Marco Mehta
9	9	11008	AW00011008	Australia	Single	Female	1975-07-04	2025-10-06	Rob Verhoff
10	10	11009	AW00011009	Australia	Single	Male	1969-09-29	2025-10-06	Shannon Carlson
11	11	11010	AW00011010	Australia	Single	Female	1969-08-05	2025-10-06	Jacquelyn Suarez
12	12	11011	AW00011011	Australia	Married	Male	1969-05-03	2025-10-06	Curtis Lu
13	13	11012	AW00011012	United States	Married	Female	1979-01-14	2025-10-06	Lauren Walker
14	14	11013	AW00011013	United States	Married	Male	1979-08-03	2025-10-06	Ian Jenkins
15	15	11014	AW00011014	United States	Single	Female	1973-11-06	2025-10-06	Sydney Bennett
16	16	11015	AW00011015	United States	Single	Female	1984-08-26	2025-10-06	Chloe Young
17	17	11016	AW00011016	United States	Married	Male	1984-10-25	2025-10-07	Wyatt Hill
18	18	11017	AW00011017	Australia	Single	Female	1949-12-24	2025-10-07	Shannon Wang
19	19	11018	AW00011018	Australia	Single	Male	1955-10-06	2025-10-07	Clarence Rai
20	20	11019	AW00011019	Canada	Single	Male	1983-09-04	2025-10-07	Luke Lal
21	21	11020	AW00011020	Canada	Single	Male	1984-03-19	2025-10-07	Jordan King

PRODUCTS TABLES

	product_key [PK] integer	product_id integer	product_number character varying (30)	product_name character varying (150)	category_id character varying (20)	category character varying (50)	subcategory character varying (50)	maintenance character varying (10)	cost numeric (10,2)
1	1	210	FR-R92B-58	HL Road Frame - Black- 58	CO_RF	Components	Road Frames	Yes	0.(
2	2	211	FR-R92R-58	HL Road Frame - Red- 58	CO_RF	Components	Road Frames	Yes	0.(
3	3	348	BK-M82B-38	Mountain-100 Black- 38	BI_MB	Bikes	Mountain Bikes	Yes	1898.(
4	4	349	BK-M82B-42	Mountain-100 Black- 42	BI_MB	Bikes	Mountain Bikes	Yes	1898.(
5	5	350	BK-M82B-44	Mountain-100 Black- 44	BI_MB	Bikes	Mountain Bikes	Yes	1898.(
6	6	351	BK-M82B-48	Mountain-100 Black- 48	BI_MB	Bikes	Mountain Bikes	Yes	1898.(
7	7	344	BK-M82S-38	Mountain-100 Silver- 38	BI_MB	Bikes	Mountain Bikes	Yes	1912.(
8	8	345	BK-M82S-42	Mountain-100 Silver- 42	BI_MB	Bikes	Mountain Bikes	Yes	1912.(
9	9	346	BK-M82S-44	Mountain-100 Silver- 44	BI_MB	Bikes	Mountain Bikes	Yes	1912.(
10	10	347	BK-M82S-48	Mountain-100 Silver- 48	BI_MB	Bikes	Mountain Bikes	Yes	1912.(
11	11	317	BK-R68R-44	Road-450 Red- 44	BI_RB	Bikes	Road Bikes	Yes	885.(
12	12	318	BK-R68R-48	Road-450 Red- 48	BI_RB	Bikes	Road Bikes	Yes	885.(
13	13	319	BK-R68R-52	Road-450 Red- 52	BI_RB	Bikes	Road Bikes	Yes	885.(
14	14	315	BK-R68R-58	Road-450 Red- 58	BI_RB	Bikes	Road Bikes	Yes	885.(
15	15	316	BK-R68R-60	Road-450 Red- 60	BI_RB	Bikes	Road Bikes	Yes	885.(
16	16	311	BK-R93R-44	Road-150 Red- 44	BI_RB	Bikes	Road Bikes	Yes	2171.(
17	17	312	BK-R93R-48	Road-150 Red- 48	BI_RB	Bikes	Road Bikes	Yes	2171.(
18	18	313	BK-R93R-52	Road-150 Red- 52	BI_RB	Bikes	Road Bikes	Yes	2171.(
19	19	314	BK-R93R-56	Road-150 Red- 56	BI_RB	Bikes	Road Bikes	Yes	2171.(
20	20	310	BK-R93R-62	Road-150 Red- 62	BI_RB	Bikes	Road Bikes	Yes	2171.(
21	21	299	FR-M94B-44	HL Mountain Frame - Black- 44	CO_MF	Components	Mountain Frames	Yes	699.(

SALES TABLE

	order_number character varying (20)	product_key integer	customer_key integer	order_date date	shipping_date date	due_date date	sales_amount numeric (10,2)	quantity integer	price numeric (10,2)
1	SO54496	282	5400	2013-03-16	2013-03-23	2013-03-28	25.00	1	25.00
2	SO54496	289	5400	2013-03-16	2013-03-23	2013-03-28	5.00	1	5.00
3	SO54496	259	5400	2013-03-16	2013-03-23	2013-03-28	2.00	1	2.00
4	SO54497	174	9281	2013-03-16	2013-03-23	2013-03-28	22.00	1	22.00
5	SO54497	280	9281	2013-03-16	2013-03-23	2013-03-28	9.00	1	9.00
6	SO54498	174	4825	2013-03-16	2013-03-23	2013-03-28	22.00	1	22.00
7	SO54498	277	4825	2013-03-16	2013-03-23	2013-03-28	54.00	1	54.00
8	SO54499	289	4286	2013-03-16	2013-03-23	2013-03-28	5.00	1	5.00
9	SO54499	246	4286	2013-03-16	2013-03-23	2013-03-28	35.00	1	35.00
10	SO54500	289	1472	2013-03-16	2013-03-23	2013-03-28	5.00	1	5.00
11	SO54500	284	1472	2013-03-16	2013-03-23	2013-03-28	35.00	1	35.00
12	SO54500	244	1472	2013-03-16	2013-03-23	2013-03-28	35.00	1	35.00
13	SO54500	166	1472	2013-03-16	2013-03-23	2013-03-28	9.00	1	9.00
14	SO54501	290	2777	2013-03-16	2013-03-23	2013-03-28	4.00	1	4.00
15	SO54501	287	2777	2013-03-16	2013-03-23	2013-03-28	33.00	1	33.00
16	SO54502	286	5302	2013-03-16	2013-03-23	2013-03-28	25.00	1	25.00
17	SO54502	290	5302	2013-03-16	2013-03-23	2013-03-28	4.00	1	4.00
18	SO54502	166	5302	2013-03-16	2013-03-23	2013-03-28	9.00	1	9.00
19	SO54502	245	5302	2013-03-16	2013-03-23	2013-03-28	35.00	1	35.00

DATA PREPARATION

Filter

The WHERE clause filters out unwanted data

Sort

The ORDER BY clause arranges your data. S

Join

The JOIN statement combines data from two or more tables.

Group

The GROUP BY clause categorizes data.

BUSINESS QUESTION



1 .Top products by revenue



2 .Top 10 customers by spending

3. Monthly revenue growth

4 .Product contribution to total revenue

5. Top 3 products in each category



6 .Repeat customer analysis

7. Customer Lifetime Value



TOP 5 PRODUCT BY REVENUE ?

```sql
SELECT
    P.PRODUCT_ID,
    PRODUCT_NAME,
    SUM(S.QUANTITY * S.PRICE) AS TOTAL_REVENUE
FROM
    PRODUCTS P
    INNER JOIN SALES S ON P.PRODUCT_KEY = S.PRODUCT_KEY
GROUP BY
    P.PRODUCT_ID,
    PRODUCT_NAME
ORDER BY
    TOTAL_REVENUE DESC
LIMIT
    5;
```

2 . TOP 10 CUSTOMERS BY SPENDING ?

```sql
SELECT
    c.customer_key,
    c.name,
    SUM(s.quantity * s.price) AS total_spending
FROM customers c
JOIN sales s
    ON c.customer_key = s.customer_key
GROUP BY
    c.customer_key,
   c.name
ORDER BY total_spending DESC
limit 10;
```

3. MONTHLY REVENUE GROWTH ?

```sql
WITH monthly_sales AS (
    SELECT
        DATE_TRUNC('month',order_date) AS month,
        SUM(quantity * price) AS revenue
    FROM sales
    GROUP BY DATE_TRUNC('month', order_date)
)

SELECT
    month,
    Revenue,
     LAG(revenue) OVER (ORDER BY month) AS previous_month_revenue,

    ROUND(
        (revenue - LAG(revenue) OVER (ORDER BY month))
        * 100.0
        / LAG(revenue) OVER (ORDER BY month )   ,
        1
    ) || '%' AS Growth_percentage

FROM Monthly_sales
ORDER BY month;
```

4 .PRODUCT CONTRIBUTION TO TOTAL REVENUE

```sql
SELECT
    P.PRODUCT_NAME,
    SUM(S.QUANTITY * S.PRICE) AS REVENUE,
    ROUND(
        SUM(S.QUANTITY * S.PRICE) * 100 / SUM(SUM(S.QUANTITY * S.PRICE)) OVER (),
        2
    ) || '%' AS REVENUE_CONTRIBUTION
FROM
    SALES S
    JOIN PRODUCTS P ON S.PRODUCT_KEY = P.PRODUCT_KEY
GROUP BY
    P.PRODUCT_NAME
ORDER BY
    REVENUE DESC;
```

5. TOP 3 PRODUCTS IN EACH CATEGORY ?

```sql
WITH
    PRODUCT_REVENUE AS (
        SELECT
            P.PRODUCT_NAME,
            P.CATEGORY,
            SUM(S.QUANTITY * S.PRICE) AS REVENUE
        FROM
            PRODUCTS P
            JOIN SALES S ON P.PRODUCT_KEY = S.PRODUCT_KEY
        GROUP BY
            P.PRODUCT_NAME,
            P.CATEGORY
    ),
    RANKED_PRODUCTS AS (select
        PRODUCT_NAME,
        CATEGORY,
        REVENUE,
        ROW_NUMBER() OVER (
            PARTITION BY
                CATEGORY
            ORDER BY
                REVENUE
        ) AS RN FROM
    PRODUCT_REVENUE)
SELECT
    PRODUCT_NAME,
    CATEGORY REVENUE
FROM
    RANKED_PRODUCTS
WHERE
    RN <= 3
ORDER BY
    CATEGORY,
    REVENUE DESC;
```

6 . REPEAT CUSTOMER ANALYSIS ?

```sql
SELECT
    C.CUSTOMER_KEY,
    C.NAME,
    COUNT(DISTINCT (S.ORDER_NUMBER)) AS ORDER_COUNT
FROM
    SALES S
    JOIN CUSTOMERS C ON S.CUSTOMER_KEY = C.CUSTOMER_KEY
GROUP BY
    C.CUSTOMER_KEY
HAVING
    COUNT(DISTINCT S.ORDER_NUMBER) > 1;
```

7. CUSTOMER LIFETIME VALUE

```sql
SELECT
    C.CUSTOMER_KEY,
    C.NAME,
    SUM(S.QUANTITY * S.PRICE) AS CUSTOMER_LIFETIME_VALUE
FROM
    SALES S
    JOIN CUSTOMERS C ON S.CUSTOMER_KEY = C.CUSTOMER_KEY
GROUP BY
    C.CUSTOMER_KEY
ORDER BY
    CUSTOMER_LIFETIME_VALUE DESC;
```

SQL SKILL USED



- JOIN
- GROUP BY
- Aggregate Functions
- CASE WHEN
- CTE
- Subqueries
- Window Functions
- LAG()
- Date Functions
- ORDER BY
- LIMIT

PROJECT CONCLUSION

This project demonstrates how SQL can be used to transform raw bicycle sales data into meaningful business insights. The analysis helped identify product performance, customer spending, revenue trends, and potential areas for business improvement.



THANK YOU

PRESENTATION 2026

